

```
#include <stdio.h>
#define N 5

int front = -1;
int rear = -1;
int queue[N];

void enqueue(int x) {
    if ((front == 0 && rear == N - 1) || ((rear + 1) % N == front)) {
        printf("Queue is full\n");
    } else if (front == -1 && rear == -1) {
        front = rear = 0;
        queue[rear] = x;
    } else {
        rear = (rear + 1) % N;
        queue[rear] = x;
    }
}

void dequeue() {
    if (front == -1 && rear == -1) {
        printf("Queue is empty\nNo element to delete\n");
    } else if (front == rear) {
        printf("The deleted element is %d\n", queue[front]);
        front = rear = -1;
    } else {
        printf("The deleted element is %d\n", queue[front]);
        front = (front + 1) % N;
    }
}

void display() {
    if (front == -1 && rear == -1) {
        printf("Queue is empty\n");
    } else {
        printf("The queue elements are:\n");
        int i;
        for (i=front;i!=rear;i=(i + 1)%N) {
            printf("%d ", queue[i]);
            printf("\n");
        }
    }
}
```

```

    }
    printf("%d",queue[rear]);
}
}

int main() {
    int ch,x;
    do{
        printf("Enter your choice\n1.enqueue\n2.dequeue\n3.display\n4.exit:");
        scanf("%d",&ch);
        switch(ch) {
case 1:
            printf("Enter element to enter:");
            scanf("%d",&x);
            enqueue(x);
            break;
case 2:
            dequeue();
            break;
case 3:
            display();
            break;
case 4:
            printf("exit..");
            break;
default:
            printf("choice out of range\n");
        }}while(ch!=4);
        return 0;
}

```

```
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:2
Queue is empty
No element to delete
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:1
Enter element to enter:1
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:2
The deleted element is 1
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:1
Enter element to enter:1
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:1
Enter element to enter:2
Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:3
The queue elements are:
1
2Enter your choice
1.enqueue
2.dequeue
3.display
4.exit:4
exit..
Process returned 0 (0x0)    execution time : 11.701 s
Press any key to continue.
|
```